

RRAM-Based Spiking Neural Network With Target-Modulated Spike-Timing-Dependent Plasticity

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Abstract—The spiking neural network (SNN) training with spike timing-dependent plasticity (STDP) for image classification usually requires a lot of neurons to extract representative features and(or) needs an external classifier. Conventional bio-inspired learning methods do not cover all possible learning opportunities, resulting in limited performance. We propose a new bio-plausible learning rule, target-modulated STDP (TSTDP), for higher learning efficiency and accuracy. We also propose an SNN architecture trainable with TSTDP using temporally encoded spikes to obtain higher accuracy and improved area efficiency without using an external classifier. Using the MNIST dataset, we have shown that the proposed design achieves an accuracy of 92%, which is up to 7% improvement compared to conventional networks of similar sizes. For providing similar accuracy, up to 75% smaller network size has been shown on top of demonstrating stronger resilience to process variations. Benchmarking on the CIFAR-10 and neuromorphic DVS gesture datasets show an accuracy improvement of up to 12.4% and 3.6%, respectively.

Index Terms—STDP, RSTDP, SNN, neuromorphic networks, memristor, RRAM, ReSuMe, TSTDP.

I. INTRODUCTION

INSPIRED by the computational and energy efficiency of the brain, spiking neural networks (SNNs) were proposed for energy-efficient spike-based information processing [1]. Synaptic plasticity in the brain is perceived to be of Hebbian type or a variation of it, which includes long-term potentiation (LTP), long-term depression (LTD), and their fusion, spike timing-dependent plasticity (STDP) [2]. Complemented with competitive dynamics through winner-take-all (WTA) and lateral inhibition, STDP enables neurons to learn unique features, solely based on the timing difference between presynaptic and postsynaptic neuronal spikes. STDP has been used for classification tasks in convolutional [3] and [4], and fully connected [5], [6], [7] SNNs. Using a simplified

rectangular STDP curve, [5] showed a single-layer SNN could achieve high accuracy on image classification but required a large network size. [6] employed more biologically plausible exponential STDP and lateral inhibition in their implementation, in which the exponential mechanism would necessitate complex hardware. A design that used inhibitory weights to jointly conduct lateral inhibition and homeostasis was proposed in [7] for an even higher performance. Generally, STDP designs face inherent limitations: few real-world problems unsupervised learning can solve, the potential need for many output neurons per class to extract features from images, resulting in a large network, degraded hardware and energy efficiency, and slower training. Moreover, a supervised external classifier is needed to identify classes extracted by the STDP layer, adding more complexity. A semi-supervised scheme which adds an STDP-based supervised layer to the unsupervised architecture of [6] was proposed in [8], but the accuracy is still similar to the unsupervised schemes.

Supervised training methods of SNNs based on spike timing were proposed to address these issues [9], [10]. The tempotron [9] minimizes a cost function in case of misclassifications, requiring complex computations and inefficiency for hardware implementation. ReSuMe [10] uses a teacher neuron spiking time as a template for output neurons to let their spike timing match with the teacher's timing. They can eliminate the above-mentioned problems of unsupervised STDP and achieve higher accuracy. Another supervised STDP study which focuses on hyperparameter optimization was proposed in [11]. However, supervised STDP-based rules often use spike-rate encoding requiring excessive spikes per neuron, which can degrade energy efficiency and slow down training. Reward-modulated STDP (RSTDP) proposed in [12] uses a global reward (punishment) signal to switch between STDP and anti-STDP (aSTDP) based on the decision of the network. The reward signal is generated by comparing the output of the network to the target signal and carries information on whether the decision is favorable or not. As the target/teacher signal is not directly given to the neurons, RSTDP provides minimal supervision of neurons. However, RSTDP cannot modify weights connected to silent target neurons, hence being unable to train them. Despite the limitations described above, a persistent demand for a local learning rule tailored to SNNs is increasing due to its potential to improve energy efficiency and classification accuracy.

In hardware SNN implementations, CMOS-based weight elements [13] offer mature technology but need multiple

Manuscript received 5 May 2024; revised 25 June 2024; accepted 19 July 2024. Date of publication 19 August 2024; date of current version 2 April 2025. This work was supported in part by the National Research Foundation (NRF) Research under Grant RS-2024-00354034 provided by the Korean Government (MSIT). Design tools for this work were supported by the IC Design Education Center (IDEC), KAIST, South Korea. This paper was recommended by Associate Editor J. Seo. (*Corresponding author: Bai-Sun Kong.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TBCAS.2024.3446177>.

Digital Object Identifier 10.1109/TBCAS.2024.3446177

transistors per synapse, which leads to a large area. Nonvolatile devices such as PCM [14] and RRAM [15] with multiple analog levels can offer high weight resolution with a lower device count. SNN implementations in [5], [7], [16], [17], and [18] used one-transistor-one-resistor (1T1R) or one-resistor (1R) cells based on PCM or RRAM as synapses and CMOS circuits as leaky integrate-and-fire (LIF) neurons. While [18] demonstrated the STDP learning capability of SNN hardware, [16] proposed a scheme to solve interference of update and inference in learning with STDP on RRAM. [5], [17] showed high accuracy in image recognition and withstood device variation. In [7], a scalable SNN leveraged memristor-based inhibitory synapses to co-implement lateral inhibition and homeostasis.

In this paper, we propose a novel bio-plausible learning rule for SNNs that can switch between different weight adjustment types for higher accuracy, faster training, and better energy efficiency. We also devise RRAM-based neuromorphic hardware adopting the proposed learning rule for higher hardware efficiency. The paper is organized as follows: Section II presents the proposed learning rule, Section III describes the SNN architecture, and Section IV covers performance evaluation in comparison with conventional works. Conclusions are given in Section V.

II. TSTDTP LEARNING RULE

Synaptic plasticity can be expressed as the strengthening or weakening of synaptic weights based on the activity of presynaptic and postsynaptic neurons. Generalized as STDP, this type of weight modification has been used to explain memory and learning of neural circuits in the brain. However, according to recent neuroscience research, STDP only is no longer enough to explain various types of learning observed in the brain [19], [20]. Among these, synaptic weights of neurons can be mediated by a target signal to induce different types of weight update policies [21]. These target-induced variations in synaptic plasticity could be the basis for a more efficient and better-performing learning rule for SNNs. Based on this neurobiological discovery, we propose a target-modulated STDP (TSTDTP) that can apply different bio-plausible plasticity rules appropriate to the underlying scenario.

In the TSTDTP rule, the *Target* signal together with the pre- and post-neuronal spike timings dictate different types of synaptic plasticity. The TSTDTP rule can let a network learn features for the following three inference results: *true firing*, *false firing*, and *false silence*. In a network with WTA competition, *true firing* signifies that the target neuron has won the competition, while *false firing* means a non-target neuron has won the competition. Since the WTA mechanism allows only one neuron to fire, a firing non-target neuron results in a silent target neuron, *false silence*. These two cases happen concurrently. The conceptual representation of TSTDTP is given in Fig. 1(a) with a simple example network of two presynaptic neurons fully connected to two postsynaptic neurons. In this example, the first presynaptic neuron (black) fires before the firing of the postsynaptic neurons (blue and orange), while the second one (purple) fires later, as shown by similarly colored spikes on the right. While the first postsynaptic neuron, *Post1* (blue), is the target neuron, it is possible that either of the two output neurons wins

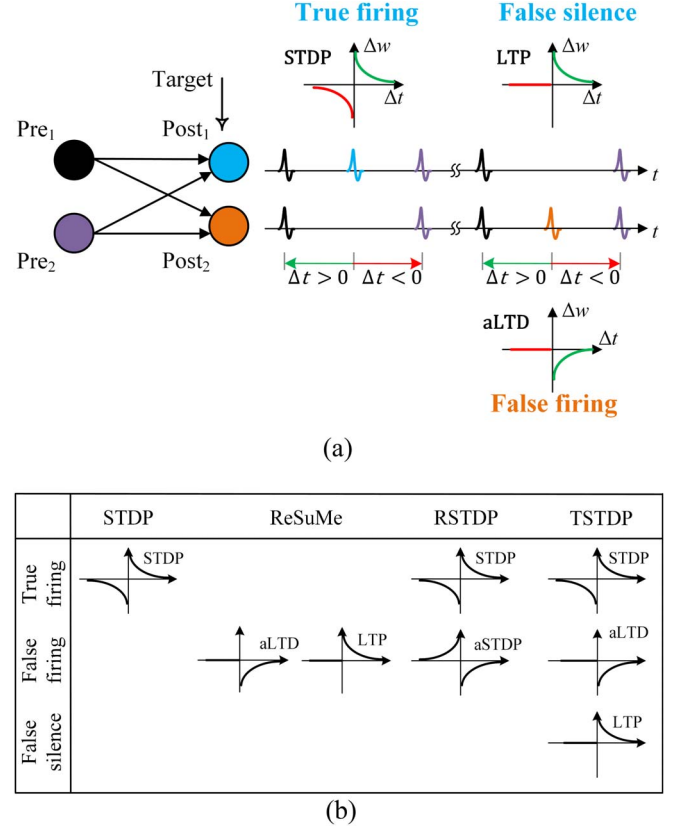


Fig. 1. TSTDTP learning rule: (a) concept of TSTDTP with a simple 2×2 network applying STDP, aLTD, or LTP. (b) Comparison with other learning principles.

the WTA competition and has a firing. These two scenarios are shown in the timing diagrams. The timing diagrams in the middle show the case in which target neuron *Post₁* is the WTA winner. In this case, since the target neuron fired and won the WTA, it denotes a *true firing*. In this case, TSTDTP lets the weights associated with *Post₁* learn in the standard STDP manner: when the pre-neuron fires (at t_{pre}) before the post-neuron (at t_{post}) firing ($\Delta t = t_{post} - t_{pre} > 0$), LTP happens, strengthening their synaptic connection. In the reverse firing order where $\Delta t < 0$, LTD takes place, weakening the connection. In the timing diagrams on the right, non-target neuron *Post₂* fires instead of target neuron *Post₁*, resulting in *false firing* for *Post₂* and *false silence* for *Post₁*, respectively. For this case, TSTDTP will reduce the connections to the *false-firing* neuron *Post₂*, and accordingly, anti-LTD (aLTD) takes place for $\Delta t > 0$, but no update for $\Delta t < 0$. For the neuron with *false silence*, *Post₁*, its weights must be increased to encourage it to win in the future. Hence, LTP takes place for $\Delta t > 0$. The firing time of the false firing neuron is used as a reference to calculate Δt for the falsely silent neuron. The general algorithm for TSTDTP is also presented in Algorithm 1 showing the specific type of weight update protocol in different inference cases. For each presynaptic neuron $i \in \{1, 2, \dots, N\}$, where N is the total number of presynaptic neurons, STDP is done when the target neuron j^* fires to update associated weights connecting postsynaptic neuron j^* and presynaptic neurons i 's with positive or negative Δt . Otherwise, aLTD is carried out for the weights connected to the false firing neuron j with positive Δt , and LTP is done for the weights to the target neuron with positive Δt .

Algorithm 1 Learning with TSTD

Input: Pre-neuron index $i \in \{1, 2, \dots, N\}$, WTA winner post-neuron index j , target neuron index j^*

Output: weight update Δw

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1: for  $i \leq N$  do
2:   if  $(j = j^*)$  then
3:      $\Delta t_{ij^*} \leftarrow t_j - t_i$  // True firing
4:      $\Delta w_{ij^*} \leftarrow STDP(\pm \Delta t_{ij^*})$ 
5:   else
6:      $\Delta t_{ij} \leftarrow t_j - t_i$ 
7:      $\Delta t_{ij^*} \leftarrow t_j^* - t_i$ 
8:      $\Delta w_{ij} \leftarrow aLTD(+\Delta t_{ij})$  // False firing
9:      $\Delta w_{ij^*} \leftarrow LTP(+\Delta t_{ij^*})$  // False silence
10:  end if
11: end for
12: return  $\Delta w$ 

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Let us compare the proposed TSTD rule having weight adjustments in the three cases described above to conventional learning rules like unsupervised STDP, RSTD, and ReSuMe, as shown in Fig. 1(b). The unsupervised STDP has no target, so firing output neurons can always be considered to be *true firing*, resulting in no weight modification for the *false firing* and *false silence* cases. In ReSuMe, the case where both the output neuron and the teacher neuron fire at the same time can be considered as a *true firing* while the cases where the output neuron fires earlier or later than the teacher neuron can be regarded as two types of *false firing* having slightly different update rules. Hence, ReSuMe can only make weight modifications in case of *false firing*, disregarding the cases of *true firing* and *false silence*. RSTD updates weights in two ways, depending on a reward/punishment signal. When the WTA winner output neuron belongs to the target class, it is rewarded. Otherwise, it is punished. These two cases can be viewed as *true firing* and *false firing*, respectively. However, RSTD ignores the *false silence* case, which may let the network be unable to learn from images that cause the target neuron to be silent. All these conventional learning rules make weight changes in up to two cases at most, posing a limitation on what they can learn. On the other hand, because TSTD can make proper weight adjustments in all three possible cases as described above, it is a comprehensive learning rule for SNNs that delivers more effective weight modifications for improved accuracy.

III. SPIKING NEURAL NETWORK WITH TSTD

A. RRAM Synapse

Our SNN adopting the proposed TSTD uses an RRAM-based 1T1R cell as a synapse. The conductance (G) of the RRAM changes with suitable programming pulses, whose characteristics are modeled based on experimental data [5]. The conductance increase and decrease are exponential functions of the current conductance:

$$\Delta G_p = \alpha_p e^{-\beta_p \frac{G - G_{min}}{G_{max} - G_{min}}} \quad (1)$$

and

$$\Delta G_d = \alpha_d e^{-\beta_d \frac{G_{max} - G}{G_{max} - G_{min}}} \quad (2)$$

where ΔG_p and ΔG_d are the amounts of conductance increase (potentiation) and decrease (depression), respectively. Parameters α_p and α_d are conductance change step sizes while β_p and β_d capture the nonlinearity in potentiation and depression, respectively. G_{max} and G_{min} are the normalized maximum and minimum conductance, with G being the current conductance.

B. Network Architecture and Operation

An $R \times C$ neuromorphic network employing TSTD is shown in Fig. 2(a), where R and C are the row and column sizes of the synaptic array. It realizes the fully connected single-layer network on the left. An input image in the form of an $R \times I$ column matrix is fed into R pre-neurons as input. Each pre-neuron connects to all the synapses in the same row and all synapses in a column are connected to a post-neuron at the bottom of the column. There are C post-neurons as outputs in the network.

The highlighted column in Fig. 2(a) is depicted in more detail in Fig. 2(b) to show the TSTD operation from an output neuron's perspective. It contains R pre-neurons and synapses, a post-neuron (Post1), and a TSTD switch (TS) block. Each pre-neuron connects to the selector transistor gate terminal in each synapse, while the post-neuron is connected to all synapses through their selector transistor source terminals. The post-neuron contains a membrane potential (V_{mp}) integrator, a comparator, and a spike generator (SG) block. The integrator accumulates the incoming postsynaptic current to membrane potential while the comparator compares V_{mp} with firing threshold V_{th} for post-neuronal firing. The SG block generates three signals: P^{STD} and P^{aLTP} for programming RRAMs in case of weight updates in the form of STDP and aLTD protocols [see Fig. 1(b)], respectively, and S_{post} for indicating the post-neuronal firing. So, the block is composed of three edge-triggered pulse generator sub-blocks that generate these pulse sequences upon the rising-edge of the input received from the comparator. In addition to P^{STD} and P^{aLTP} , there is a third programming signal for LTP type weight update, P^{LTP} , generated by the pulse generator (PG) block in Fig. 2(a). S_{post} 's from all post-neurons are fed into the WTA block in Fig. 2(a) to select the winning neuron. A WTA circuit proposed in [22] can be used for this purpose. Signal $mpReset$ generated by the WTA block resets V_{mp} of all output neurons. In addition, the PG block also generates F_{frwd} signal, which is used to drive the post-synaptic current for membrane potential integration by providing a small negative voltage V_{frwd} . The WTA winning neuron is indicated by turning on the corresponding bit in the C -bit wide *Winner* bus output from the WTA block. The output neurons are equally assigned between classes in prior i.e., there are W/N_C neurons per class, where N_C is the number of classes in the dataset and C should be an integer multiple of N_C . Similarly, the target class is indicated by turning the corresponding line ON from the N_C -bit wide *Target* bus. The PG block compares whether the winner neuron is a target neuron or not to identify the resulting inference type. When the corresponding Target line is ON for the

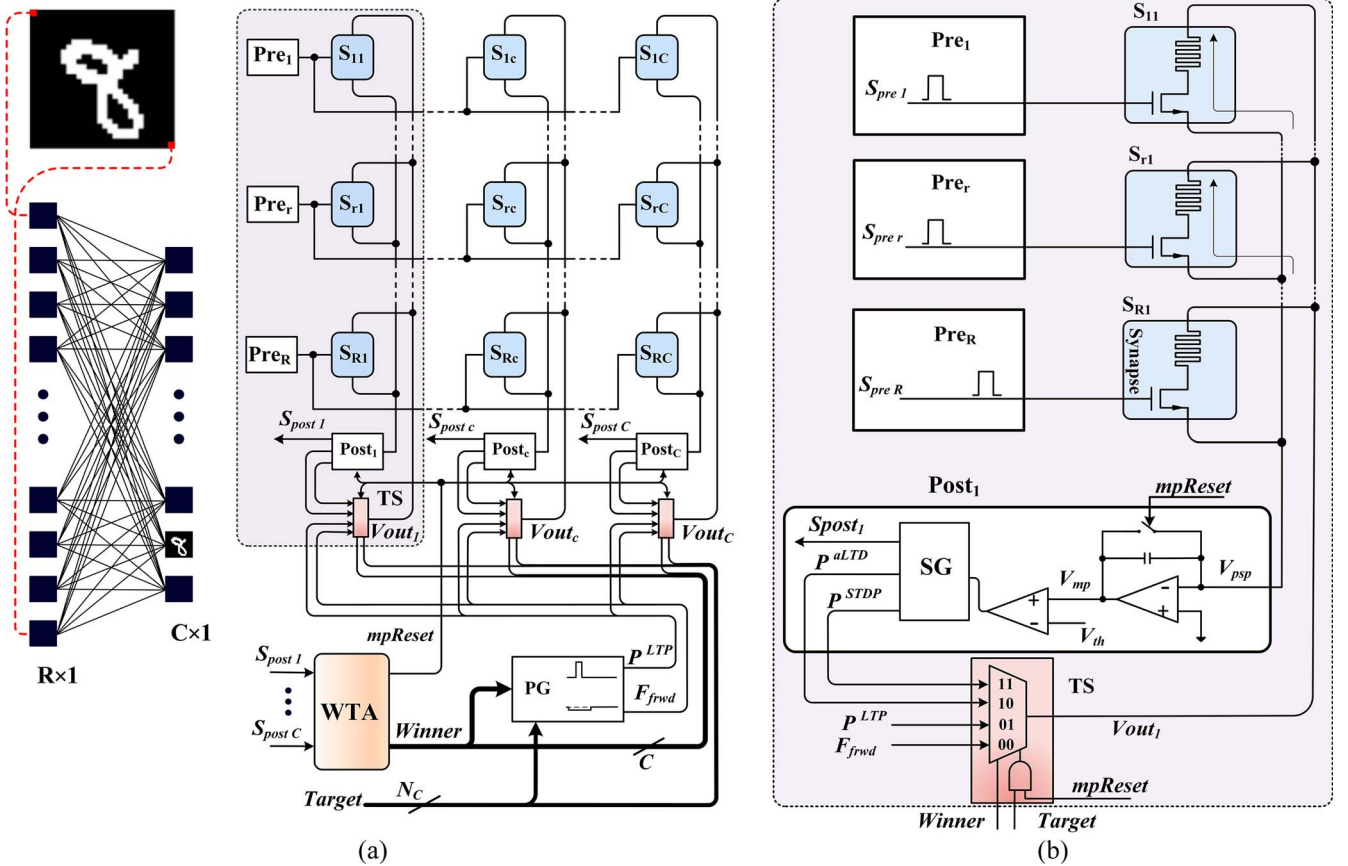


Fig. 2. Proposed SNN architecture employing TSTD learning rule: (a) $R \times C$ overall network, (b) structure of a column highlighted in (a).

winning neuron, PG identifies the inference case as *true firing*. When the corresponding *Target* line is OFF, the instance would be identified as *false firing/false silence*, and a programming pulse is released on P^{LTP} . The TS block, which is a 4-to-1 multiplexer with an AND gate, uses *Winner* and gated *Target* Signals to select one among F_{frwd} for forward propagation, and $P^{STD P}$, P^{LTP} , or P^{aLTD} for weight updates and pass it to V_{out} . *Target* is gated by *mpReset* as it is information is used only during weight update.

The conceptual timing diagrams in Fig. 3 illustrate the operation of the proposed network. Spikes are temporally encoded into two timesteps in a period (T). When an input pixel is active (ON), the associated pre-neuron fires at the start of the first time step, whereas when it is inactive (OFF) the pre-neuron fires at the start of the second time step (top two waveforms in Fig. 3 left). Spikes from all pre-neurons with ON pixels, represented as $Spre_{ON}$ activate the selector transistors in corresponding synapses in Fig. 2(b). Since signals *Winner* and *Target* after gated with *mpReset*, are set to zero at the start of the inference, V_{out} follows F_{frwd} until a post-spike occurs at t_{post} . Since the signal supplies a small negative voltage and V_{psp} is virtually grounded, an upward current flows through all ON-synapses for the duration of the presynaptic pulses or until a post-spike occurs, which causes V_{mp} to increase. When V_{mp} reaches V_{th} at t_{post} , the SG block issues a post-neuronal spike on S_{post} (indicating the output neuron has fired) along with RRAM programming pulses: A set pulse at an amplitude of V_{set} followed by a reset pulse at an amplitude of V_{reset} on $P^{STD P}$, and a reset pulse on P^{aLTD} (Fig. 3

left). Then, V_{mp} of all output neurons is reset to 0 V upon the arrival of *mpReset* from the WTA block. The block also issues an appropriate voltage on the WTA bus to indicate the winner neuron. Depending on the values of *Winner* and *Target*, one of the three programming signals is passed to V_{out} , altering the RRAM conductance upon coincidence with presynaptic pulses. The *true firing* case is depicted in Fig. 3 left. For a target neuron, the corresponding line in *Target* bus is ON, as shown in the figure. When the target neuron wins the WTA competition, both *Winner* and *Target* will be 1, so V_{out} will follow $P^{STD P}$.

The set pulse on V_{out} coincides with spike $Spre_{ON}$, increasing the conductance of the corresponding synapses, $G_{h_{ON}}$; depicting LTP in STDP [Fig. 1(b)] as $\Delta t_{ON} = t_{post} - t_{pre_{ON}} > 0$. Individual conductance change is possible (only one ON pixel in $Spre_{ON}$ but here we used column-wise changes for parallelism. Since the $Spre_{OFF}$ spike occurs in the second time step in a period, after the postsynaptic spike, and overlaps with the reset pulse, the corresponding conductance, $G_{h_{OFF}}$ decreases. With the negative timing difference, $\Delta t_{OFF} = t_{post} - t_{pre_{OFF}} < 0$, it constitutes LTD. When a non-target neuron wins the WTA, the *false firing* case is resulted in for the non-target winner neuron since it should not fire, and the *false silence* case for all neurons in the target class since they fail to win the competition. Fig. 3 right-top shows (with some waveforms already in Fig. 3 left redacted to avoid redundancy) the resulting aLTD weight update in the case of *false firing*, which is represented by *Winner* = 1 and *Target* = 0 where V_{out} follows P^{aLTD} until the period ends.

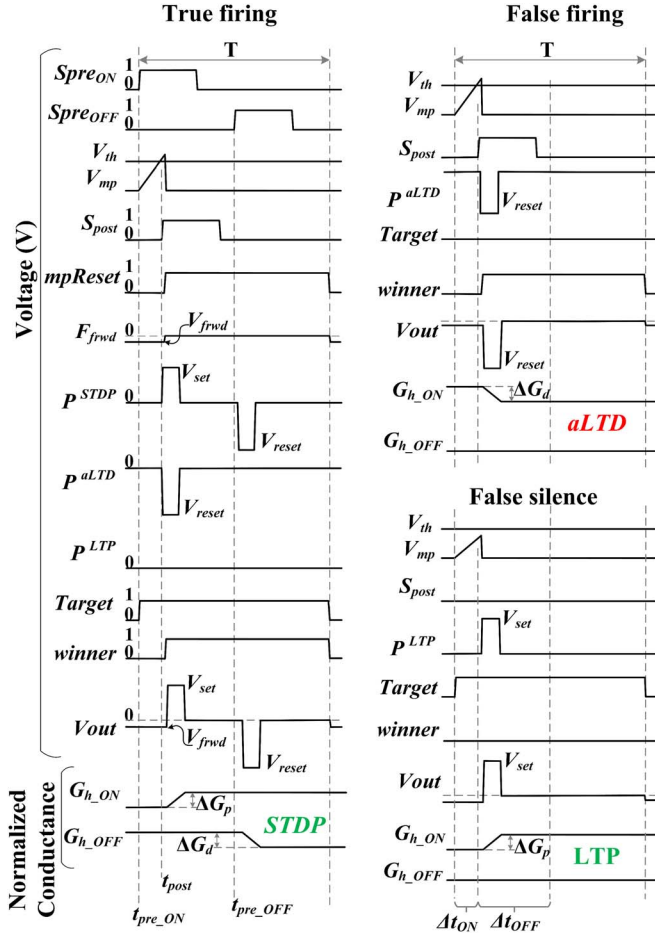


Fig. 3. Timing diagrams for true firing (left) and false firing/false silence (right).

The $Spre_{ON}$ spike overlaps with the reset pulse, which makes G_{h_ON} decrease, i.e., aLTD. In the case of false silence, P^{LTP} is passed on to V_{out} . Thus, as shown in Fig. 3 right-bottom with $Winner = 0$ and $Target = 1$, the overlap of the set pulse in V_{out} with $Spre_{ON}$ will result in LTP, increasing G_{h_ON} . If the duration of the overlap between the $Spre_{ON}$ and set pulses is shorter than the programming pulse's duration, the amount of conductance change will be smaller; otherwise, executing step-function like STDP for all the amount of Δt .

IV. PERFORMANCE EVALUATION

We trained the proposed neuromorphic networks of different sizes on the modified National Institute of Standards and Technology (MNIST) dataset [23], a ten-class handwritten digit database containing 60,000 training and 10,000 testing images of size 28x28. The images were binarized and reshaped to column vectors, as seen in the left part of Fig. 2(a). The proposed network was implemented with a custom MATLAB code. The conductance of synapses was randomly initialized with a normalized conductance mean of 0.8 and a standard deviation of 0.1. Values of parameters α_p , α_d , β_p , and β_d are set to 0.01, 0.005, 3, and 3, respectively, and G_{max} and G_{min} are normalized values and are set to 1 and 0, consistent with the values used in [5]. With only 10

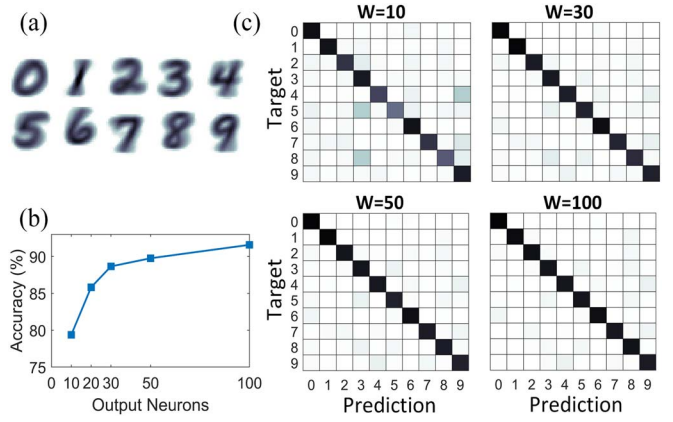


Fig. 4. Proposed SNN with TSTD on MNIST dataset: (a) learned weight conductance of RRAM synapses in the network for $N = 10$, (b) accuracy (%) for variable number of output neurons, (c) confusion matrix for $N = 10, 30, 50$ and 100.

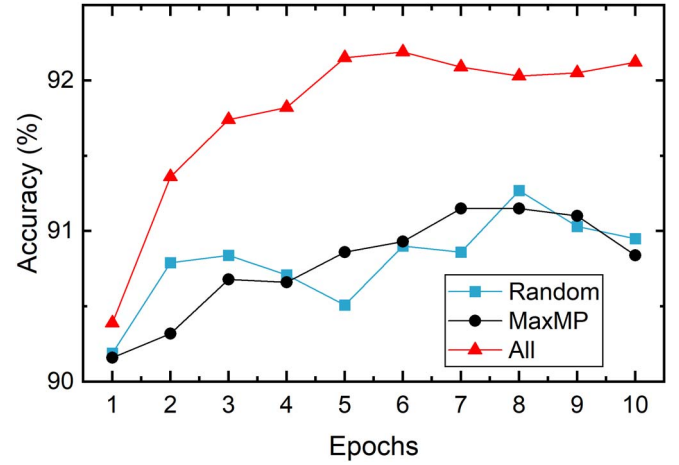


Fig. 5. Accuracy of different ways of selecting adapting false silent neurons.

output neurons, one for each class, a test accuracy of 79.5% was achieved, much higher than the 60% test accuracy obtained by the original STDP [5]. The features extracted by the network for the case where there are just ten output neurons are shown in Fig. 4(a), which indicates that, even with ten output neurons, TSTD can well extract representative features relevant to each class. Meanwhile, due to its unsupervised nature, the original STDP may consider different handwritings of a number to belong to other classes of numbers, which makes it hard to consistently achieve high accuracy with just ten output neurons. For TSTD, accuracy rises with more output neurons, achieving up to 92% with 100 output neurons as shown in Fig. 4(b). The confusion matrices shown in Fig. 4(c) illustrate a reducing confusion trend as the number of output neurons increases.

To enhance the accuracy of the TSTD rule, we explored strategies for selecting single or multiple silent neurons in the target class for learning in the false silence case. We tested three cases: single random selection, the highest V_{mp} (MaxMP) selection, and all-neuron selection. Selecting all neurons in the class yields the best accuracy (Fig. 5), as it grants all neurons in the class a chance to learn features from the image. In contrast,

TABLE I
ACCURACY AND NETWORK COMPLEXITY COMPARISON OF NEURAL NETWORKS

	Learning Rule	Data Encoding	External Classifier	Network Size	RRAM Count	Normalized RRAM Area*	Accuracy (%)
[5]	STDP	Rate	Required	784×100	78,400	1	85
				784×300	235,200	3	93
[6]	STDP	Rate	Required	784×100	78,400	1	82.9
				784×400	313,600	4	87
[7]	STDP	Rate	Required	784×100	78,400	1	87
				784×400	313,600	4	91.7
[10]	ReSuMe	Temporal	Not required	784×100	78,400	1	89.31
				784×200	156,800	2	90.29
[12]	RSTDP	Temporal	Not required	784×100	78,400	1	90.4
				784×200	156,800	2	91.45
This work	TSTDP	Temporal	Not required	784×100	78,400	1	92

*Normalized to the total device area of the proposed network.

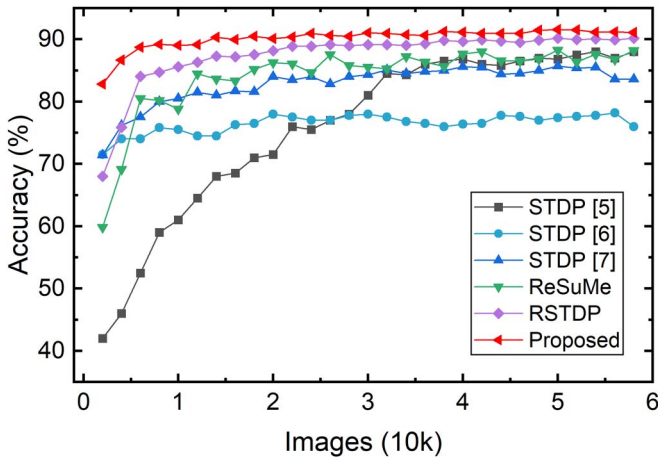


Fig. 6. Learning efficiency of this work in comparison with others.

MaxMP focused on the closest feature, yielding accuracy similar to the random selection. Moreover, these latter approaches may require a random number generator or a V_{mp} comparison and selection hardware.

We compared the accuracy and network complexity of TSTDP with those of conventional works in [5], [6], [7], [10], and [12] in Table I. Designs in [5], [6], and [7] use rate-based encoding, and unsupervised STDP, requiring supervised external classifiers to categorize the features learned by STDP. ReSuMe [10] employs rate-based encoding with supervised STDP and spiking rate-based prediction, while RSTDP [12] uses temporal coding and first-to-spike-based categorization. Both rules do not require external classifiers. Our scheme employs temporal encoding with only one spike per image for each neuron with no external classifier. Hence, it is highly sparse and possesses faster learning and energy efficiency. For a fair comparison in Table I, both ReSuMe and RSTDP learning rules were re-implemented with temporal encoding with only one spike per image as ours does. As seen in Table I, for an identical network of 784×100 (100 output neurons) the proposed network achieves the highest accuracy among all designs with up to

9.1% improvement. Compared with the recent work in [8] that rather uses software weights, the proposed work has a 3.44% accuracy improvement. Another important aspect of TSTDP we can observe from Table I is that it requires a smaller network size to provide similar accuracy, which translates to reduced power consumption. The result indicates up to four times smaller synaptic device count compared to conventional designs, implying reduced energy and area consumption.

Our network also shows higher learning efficiency, which is a measure of how fast it learns. With 100 output neurons (Fig. 6), the proposed network needs a smaller number of training images to reach its highest accuracy, while conventional ones require a larger number of images. This is because TSTDP executes effective weight modification for three possible scenarios for efficient feature extraction.

To assess the effect of device-to-device variation of RRAM on accuracy, we trained a 784×100 network on MNIST for device parameter variations of 0%, 15%, 25%, and 35%, defined as the ratio of standard deviation to the mean value. Variation is added to each device parameter at the start of a simulation and is maintained throughout the simulations, defining the characteristics of each device slightly (depending on the amount of variation) different from the others. These variations do not need to be sampled for every weight update, but they generally, and the effect of variation persists throughout the training process. For each variation level, we repeated the simulation 100 times and calculated the mean values of the recognition accuracy. Fig. 7 shows the individual and combined effects of variation in set conductance (G_{max}), reset conductance (G_{min}), conductance change step sizes (α_p and α_d), and constraint in memristor conductance states of RRAMs. The inset shows the effect of limiting the number of conductance states (levels) in the RRAM on the network's accuracy. With 1000 levels, the RRAM has sufficient states to make fine conductance adjustments during the training process, and the network achieves its highest accuracy. The network does not exhibit a significant accuracy drop for as few as 100 conductance levels. When the levels are limited to 60, the network shows an accuracy drop of 0.6%. As seen in Fig. 7, the accuracy is negligibly degraded for variations in G_{max}

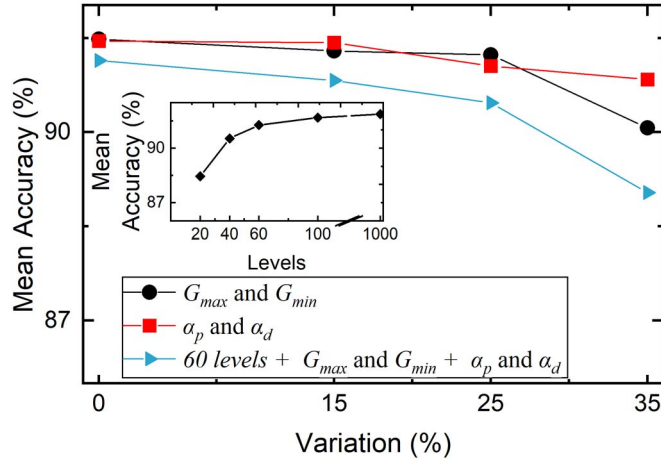


Fig. 7. Tolerance to variations on device parameters G_{max} , G_{min} , α_p , α_d , and the number of conductance states of the proposed network.

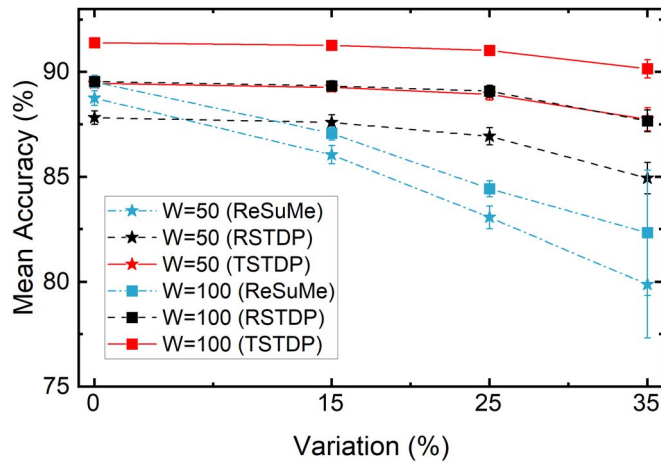


Fig. 8. Comparison of tolerance to device-to-device variation among neural networks.

and G_{min} up to 25%. The network shows an accuracy level above 90% up to 35% variation in conductance change step sizes α_p and α_d . The figure also shows, with 60 conductance levels, the cumulative effect of G_{max} and G_{min} , and α_p and α_d variations on the accuracy. As expected, the cumulative effect in variation of different parameters shows a larger accuracy drop than the individual variation of parameters. For the comparison of our design's tolerance to variation with other designs, the recognition accuracies of 784x50 and 784x100 networks with variations are depicted in Fig. 8. The error bars represent one standard deviation and are very small at low variations. The proposed network exhibits the largest tolerance for both network sizes compared to other networks. The robustness comes from its ability to learn for all possible inference cases to extract well-representative features even under non-ideal device characteristics. At 35% variation, the proposed network sees a mere 1.33% (1.77%) accuracy drop for 100(50) output neurons.

In Addition to the MNIST dataset, we have also evaluated the performance of the proposed design on the DVS gesture [24] as a neuromorphic spatiotemporal dataset and the CIFAR-10 [25] as a more complex dataset. The DVS gesture dataset is a

TABLE II
PERFORMANCE EVALUATION ON DVS GESTURE AND CIFAR-10 DATASETS

Dataset	Item	Network	Accuracy (%)
DVS gesture	[26]	16384×100	46.82
	This work	16384×100	50.42
CIFAR-10	STDP	3072×100	27.74
	RSTDp	3072×100	24.24
	This work	3072×100	35.63

spatiotemporal neuromorphic dataset created by recording 29 subjects' hand gestures on a DVS camera of 128×128 pixels. Ten different hand gestures, such as clapping, right/left hand waving, hand roll, drums, and air guitars, are repeated by each user, and each user makes one random gesture that is not repeated by others. The data from the first 23 users are for training, while the remaining data is for testing. Similar to [26], the eleventh random gesture class is excluded for simplicity. The dataset was presented to the proposed scheme with each gesture segmented into 100 timesteps. The results in Table II show on a network of 16384×100 , the proposed scheme reaches an accuracy of 50.42%, which is 3.6% higher than the work in [26] on a similar network size. The result demonstrates the ability of the proposed learning rule to work with authentic neuromorphic datasets. It also demonstrates the ability of TSTDp to work on data with multiple time steps, not just binary. The CIFAR-10 is a far more complex dataset of 32×32 color images belonging to diverse classes such as cars, boats, animals, planes, etc. It allocates 50,000 images for training and 10,000 images for testing. To account for the three channels of color and 32x32 dimension of images, we used a network of size 3072×100 . Unlike in the MNIST case, where the images were binarized and two timesteps were used, ten timesteps are used in this case, where intensities of the pixels are grouped into 10 to create intensity-to-latency coding of the images. The pixel intensities in the same group spike on the same time step, and the higher the intensity the earlier they fire. Table II compares the performance of the proposed TSTDp rule on the CIFAR-10 dataset with that of unsupervised STDP and RSTDp, which are also implemented with temporal encoding and only one spike per neuron per image for a fair comparison. TSTDp reached an accuracy of 35.63%, outperforming both STDP (27.24%) and RSTDp (24.24%) learning rules. The lower accuracy of RSTDp here is due to the reduced effectiveness of aSTDp, specifically the aLTP component, in its role of discouraging false firing neurons. The CIFAR-10 dataset contains significantly diverse images even within a class, making it challenging to distinguish the most common foreground and background regions in the images. This difficulty seems to have hindered the functionality of aSTDp, ultimately reducing the accuracy. The result here shows the potential of TSTDp to learn on non-binarized complex datasets.

V. CONCLUSION

In this paper, we presented a new learning rule for SNNs and its implementation with RRAM synapses. We have shown, using the MNIST dataset, that the proposed network can achieve the highest accuracy compared with existing conventional

works, eliminates the need for an external classifier, and can reduce network complexity by half or more. Furthermore, with one spike per neuron, it can provide high sparsity resulting in energy and hardware efficiency, which can be exploited in complex neural networks for improving accuracy from an algorithmic point of view and for designing efficient SNN accelerators from a hardware point of view. The proposed scheme has also demonstrated improved performance on more complex and spatiotemporal datasets.

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